

Problem

Design a problem to help other students better understand how to find the exponential Fourier series of a given periodic function.

Step-by-step solution

Step 1 of 4

Let us consider a periodic function,

$$f(t) = \frac{At}{2} \quad 0 < t < T$$

Since the period T

$$\omega_0 = \frac{2\pi}{T}$$

The exponential Fourier series coefficients

$$C_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$C_0 = \frac{1}{T} \int_0^T \frac{At}{2} dt$$

$$C_0 = \frac{A}{2T} \left[\frac{t^2}{2} \right]_0^T$$

$$C_0 = \frac{A}{4T} [T^2 - 0]$$

$$\therefore C_0 = \frac{AT}{4}$$

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$$C_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{T} \int_0^T \frac{At}{2} e^{-jn\omega_0 t} dt$$

$$C_n = \frac{A}{2T} \int_0^T t e^{-jn\omega_0 t} dt$$

$$C_n = \frac{A}{2T} \left[\frac{te^{-jn\omega_0 t}}{-jn\omega_0} \right]_0^T - \int_0^T \frac{e^{-jn\omega_0 t}}{-jn\omega_0} dt$$

$$C_n = \frac{A}{2T} \left[\frac{T e^{-jn\omega_0 T}}{-jn\omega_0} - \frac{e^{-jn\omega_0 t}}{(-jn\omega_0)^2} \right]_0^T$$

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$$C_n = \frac{A}{2T} \left[\frac{T e^{-jn2\pi}}{-jn \frac{2\pi}{T}} - \frac{e^{-jn \frac{2\pi}{T} T} - 1}{\left(-jn \frac{2\pi}{T}\right)^2} \right]$$

$$C_n = \frac{A}{2T} \left[\frac{T e^{-jn2\pi}}{-jn \frac{2\pi}{T}} - \frac{e^{-jn2\pi} - 1}{\left(-jn \frac{2\pi}{T}\right)^2} \right]$$

We know that $e^{-jn2\pi} = 1$

$$C_n = \frac{A}{2T} \left[\frac{T^2}{-2jn\pi} + \frac{1-1}{\frac{n^2 4\pi^2}{T^2}} \right]$$

$$C_n = \frac{jTA}{4n\pi}$$

Step 4 of 4

Therefore the exponential Fourier series

$$f(t) = C_0 + \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$f(t) = \frac{AT}{4} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{jTA}{4n\pi} e^{jn\omega_0 t}$$